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Widespread drought-driven declines in streamflows and water quality in the Upper Colorado River Basin during 1998–2022



Emily Nagamoto ^{1,2}, Mohammed Ombadi ^{1,3,7}, Fabio Ciulla^{1,7}, Jared D. Willard^{4,5}, Rosemary W. H. Carroll ⁶ & Charuleka Varadharajan ¹

The cascading effects of meteorological droughts on river flows and water quality in mountainous regions are poorly understood. Using a data-driven framework, we studied drought impacts on streamflow, water temperature (WT), specific conductance (SC) in the Upper Colorado River Basin (1998–2022). Severe meteorological droughts in 2001–2002, 2012, 2018, 2020–2021 reduced annual streamflows by 15%, increased WT by 9% and SC by 5% relative to normal adjacent years. Declines were greatest in spring and summer, with reduced peak flows. Streamflow and SC recovered until 2018, after which successive droughts prolonged impacts, whereas WT effects were transient. Catchments with higher freshwater withdrawals, deciduous forest cover, air temperatures, residence times, finer soils had greater streamflow declines, while higher subsurface flows and infiltration enabled resilience. WT increases were associated with higher elevation, slopes, precipitation and lower agriculture, shrub cover. Open water and reservoirs mitigated SC increases. Thus, catchment attributes influence drought vulnerability in mountainous regions.

Meteorological droughts are periods with abnormally low water availability that occur due to low precipitation and can be compounded by elevated air temperatures^{1–3}. Meteorological drought can propagate into hydrological (reduced terrestrial water availability), agricultural (soil moisture deficits affecting crops), and socio-economic (water demand exceeding supply) droughts, which can considerably impact ecosystem health and human societies^{2,4–6}. For example, between 1980 and 2022, drought events in the United States (US) caused severe economic losses of more than 325 billion USD⁷. The frequency, severity and duration of meteorological drought events is predicted to increase due to climate change^{7–11}.

One of the primary consequences of meteorological drought is reduced stream flows, which have been observed in many regions around the world^{12–16}. This can occur for various reasons including reduced recharge from decreased precipitation², higher temperatures leading to increased evapotranspiration^{12,17,18}, decreased groundwater or soil moisture storage^{15,16,19}, and anthropogenic activities such as increased surface water

abstraction or groundwater pumping²⁰. The degree to which streamflows decline is governed by a combination of climatic factors (e.g., temperature, precipitation) as well as biophysical (e.g., topography, geology, vegetation) and anthropogenic (e.g., land use, reservoir management) catchment attributes^{12–15,21}. Decreases in streamflow can have cascading impacts on water quality and ecosystem health^{2,22–26}. For example, declines in water quality have been observed during drought events with increased water temperatures^{24,26,27} and salinity^{25,26}. Changes in river thermal or salinity regimes can have negative effects on local and downstream ecosystems and aquatic life²⁸. Thus, analyses of catchment vulnerability to drought, and corresponding water management decisions to mitigate negative impacts, should consider the combined effects of changes in streamflow and water quality.

Although there are numerous studies on the impacts of drought in different types of watersheds, we still lack a clear understanding of how meteorological droughts affect river flows and water quality in mountainous

¹Earth & Environmental Sciences Area, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ²Department of Geography, University of Colorado Boulder, Boulder, CO, USA. ³Department of Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI, USA. ⁴Computing Sciences Area, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ⁵Artificial Intelligence (AI), Learning, and Intelligent Systems Group, National Laboratory of the Rockies, Golden, CO, USA. ⁶Division of Hydrologic Science, Desert Research Institute, Reno, NV, USA. ⁷These authors contributed equally: Mohammed Ombadi, Fabio Ciulla. e-mail: cvaradharajan@lbl.gov

regions. Mountains are ‘water towers of the world’, contributing approximately 50% of streamflows globally²⁹. Mountain systems have distinct hydrological processes that mediate streamflow and water quality compared to lowland regions. In particular, snow is the primary form of precipitation, with the majority of streamflow derived from snowmelt-driven runoff and groundwater contributions^{30–32}. Stream temperatures are influenced by air temperature as well as melting snow and glacier ice³³, groundwater³⁴, and reservoir releases³⁵. In addition, mountainous regions have steep terrain with large elevation, climatic, and vegetation gradients, and small lower-order streams, leading to substantial spatial variability in flows and water quality. Headwater catchments are frequently managed with reservoirs to regulate downstream water availability and quality. Thus, warming temperatures and shifting precipitation patterns make these regions particularly susceptible to drought, posing uncertain risks to downstream water users.

The southwestern US, and particularly the mountains in the Colorado River Basin (CRB), has been identified as a region susceptible to increased risk of drought⁷. The CRB is a critical source of water for over 40 million people in seven states, numerous tribal nations, and northern Mexico³⁶. Snowmelt from headwater catchments in the Upper Colorado River Basin (UCRB) supplies ~90% of the river flows for the entire basin^{37,38}. The UCRB has experienced historically dry and warm conditions since 2000^{18,39,40}, which combined with precipitation deficits and consequent reductions in snowpack have led to reduced streamflow and changes in the timing of snowmelt^{17,18,41–45}. Several modeling studies have shown that future warming in the UCRB will lead to earlier snow melt, increased evapotranspiration losses, decreases in snowpack and groundwater, and consequently lower streamflows, which could lead to irreversible aridification of the region^{18,32,42,43,46–51}. However, the moderate precipitation deficits since 2000 only partially explain the sharp declines in streamflows⁴¹. The reservoirs in the basin regulate river flows, holding snowmelt and baseflow during high-flow periods^{52,53}, and releasing water during low-flow seasons. With recent droughts, the ability of these reservoirs to sustain streamflows under future conditions is in doubt⁵⁴. The severe decline in flows over the last two decades and the potential long-term shifts in hydroclimatic conditions^{55,56} have necessitated changes to governance and water apportionment across the entire CRB. Although stakeholders in the UCRB consider water temperature and salinity as important water quality parameters⁵⁷, these are monitored at fewer sites and remain understudied. A couple of studies found elevated salinity⁵⁸ and water temperature⁵⁹ levels at several sites during the severe drought in 2002. A recent study found that climatic conditions only explain about 54% of the variability in streamflows and 40% of the variability in salinity, with the remaining drivers being anthropogenic activities or other watershed processes from 1986 to 2021⁶⁰. Thus, there is a general lack of understanding of the effects of drought on water quality in the basin. As a heavily managed basin that supplies critical water resources, the UCRB exemplifies the challenges mountainous basins face during drought.

The natural and human factors that affect the vulnerability of catchment flows and water quality to meteorological drought remain poorly understood, particularly in mountainous regions. To address this gap, we investigated the impacts of severe meteorological drought episodes (henceforth referred to as drought) from 1998 to 2022 on streamflows (Q), water temperature (WT), and salinity (SC) in catchments with different properties in the UCRB. Our objectives were to (1) quantify the magnitude of changes in Q, WT, and SC to droughts relative to adjacent normal years (henceforth referred to as catchment response to drought), (2) evaluate catchment recovery following drought episodes, and (3) identify catchment attributes that influence spatial variability in responses to drought.

We have developed a robust data-driven framework, DIRE (Drought Impacts and Resilience Evaluation) to identify catchment response to specific drought episodes. Our analysis for this study utilizes long-term stream monitoring data and a comprehensive dataset of 250 catchment characteristics (Geospatial Attributes of Gages for Evaluating Streamflow, GAGES-II) comprising human (e.g., land use, water management) and natural factors (e.g., climate, topography, vegetation) from the United States Geological Survey (USGS). We also use an advanced deep learning model to

predict stream temperatures in unmonitored regions to address data limitations. The DIRE framework first applies an event-based metric to quantify annual and monthly changes in Q, WT and SC (collectively referred to as response variables) during meteorological drought episodes. It then uses a combination of statistical methods to assess the persistence and recovery of these variables from drought impacts. Finally, the framework applies statistical methods, ensemble machine learning (ML) models, explainable artificial intelligence (XAI), and network science to identify the most important catchment attributes that influence changes in the response variables during drought. This modular, data-driven approach provides insights into mountainous watershed response to drought that are broadly relevant beyond the UCRB to other regions around the world.

Results

Basin-wide severe drought episodes

Severe drought episodes with basin-averaged Standardized Precipitation Evapotranspiration Index (SPEI) values less than -1 (see Methods, “Identification of meteorological drought”) occurred in 2001–2002, 2012, 2018, 2020–2021, where consecutive drought years were treated as a single event (Fig. 1 and SI Table S.1). These were dry years regionally across the UCRB, affecting both high- and low-elevation sites. Annual precipitation (P) during these drought years was consistently below the climatological average (SI Table S.2), with median declines of 23% across all sites and slight variations between events (Fig. 2A). The 2018 and 2020–21 droughts were considered separate events, as 2019 was a normal year for most sites based on the SPEI values, and some sites even had abnormally wet conditions (Fig. 1). Seasonal patterns were evident, with greater declines in May and June at most sites (Fig. 2A and Table S.2), reflecting typical weather in the region with winter snowfall and late summer monsoons. Air temperatures during drought were mostly above the climatological average (median increase of 25% across all sites) (Fig. 2B). On average, spring (April–June) air temperatures are elevated during drought years ($\sim 35\%$ relative change across all sites; Table S.2) compared to winter (January–March, -18%), summer (July–September, 5%) and fall (October–December, -48%). We note that air temperatures were low in late fall and winter (close to zero), and hence minor differences in values resulted in large percentage changes.

Drought Impacts

Most sites (170 out of 202, including 24 pristine catchments) experienced declines in annual streamflow (Q = discharge normalized by basin area) during drought years (Fig. 3A and Supplementary Information SI Fig. S.1). The largest decrease in relative change was approximately 97%, with median declines of approximately 15% across all sites (Table 1). Of the 32 sites with increased flows during drought, 30 had changes between 0% to 62%, while 2 exceeded 100%. These sites had lower median flows across the study period ($0.002 \text{ m}^3 \cdot \text{s}^{-1} \cdot \text{km}^{-2}$, $n=32$) compared to sites that experienced flow declines ($0.007 \text{ m}^3 \cdot \text{s}^{-1} \cdot \text{km}^{-2}$, $n=170$, SI Fig. S.2). Peak flows (95th percentile) were most affected, with 199 sites experiencing declines up to 99%, and a basin-wide annual decline of 47% (Table 1 and SI Fig. S.3). Low flows (5th percentile) decreased at 161 sites, with a maximum drop of 97%, and a basin-wide decrease of 15%. Most of the declines were due to lower peak flows in June ranging from 51% (2001–2002) to 78% (2012, 2018), with gradual recovery of flows in later months (Fig. 2C and SI Fig. S.4). On average, the greatest declines during drought occurred in the summer (54% from July–September), and modest declines also occurred in the spring (11% from April–June).

Most sites (29 out of 36) experienced higher WT during drought, with a maximum increase of 28% and a basin-wide median increase of about 9% (Fig. 3B and Table 1). Of the 7 sites with declining WT, 6 had decreases between 0% and 5%, and one declined by 23%. Sites that experienced WT declines during drought had a slightly higher annual median WT (10°C) than sites with WT increases (9°C). Declining sites also had a narrower range of annual mean WT ($6\text{--}11^\circ\text{C}$) than the 29 sites with increases ($4\text{--}14^\circ\text{C}$; SI Fig. S.2). Basin-wide, peak (95th percentile) and low (5th percentile) annual WT increased by about 2%–5% during drought (Table 1).

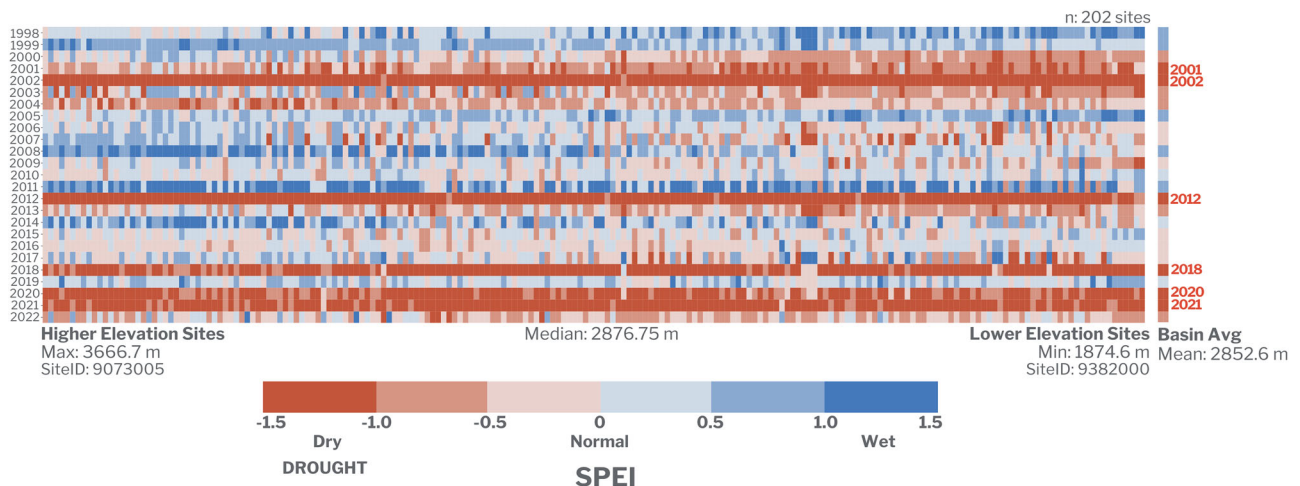


Fig. 1 | Annual average SPEI values for water years 1998–2022. The X-axis represents 202 sites across the basin in order of descending elevation from left to right. Water years are shown in the Y-axis. The right-most column shows the basin-wide average SPEI values. Average SPEI values less than -1 indicate a year of

drought (darkest red color), while SPEI values between 1 to -1 are considered normal 'reference' years (light blue and red colors). The SPEI scale is limited from -1.5 to 1.5 to highlight the difference between drought and reference years.

Seasonally, increases during drought peaked in summer (typically June), ranging from 17% to 28%, quickly recovering to reference levels by August (Fig. 2D right panel and SI Fig. S.8). Fall and winter changes were more variable across years. Unlike streamflow, WT changes were not consistent across droughts (SI Fig. S.1B), possibly due to differences in sites with data available across events (SI Table S.3, SI Fig. S.6). Water temperatures were below average in earlier droughts (2001–2002, 2012) and mostly above average in recent years (2018, 2020–2021), especially in spring and summer (Fig. 2D, left panel). The exceptionally warm temperatures in November 2018 were likely driven by warm air temperatures during that period (Figs. 2b and d, right panels). The higher temperatures in 2020–2021 were mostly caused by differences in WT of sites with available data, since many of the sites with warmer temperatures in 2021 did not have data for the earlier periods (SI Fig. S.7).

Most sites (16 out of 23) had higher SC during drought (Fig. 3C), with a maximum annual median increase of 45% and a basin-wide average increase of 5% (Table 1). Seven sites experienced declines, with a maximum drop of 11%. The sites experiencing declines had a higher annual median salinity during the study period ($862 \mu\text{S}\cdot\text{cm}^{-1}$) compared to sites with increases during drought ($730 \mu\text{S}\cdot\text{cm}^{-1}$; SI Fig. S.2). During drought, monthly median SC values were lower than the 1998–2022 average (Fig. 2E, left panel). Yet, relative changes of SC increased starting in May and remained positive through September during drought (Fig. 2E, right panel and SI Fig. S.8), with a notable increase of 86% in June 2018. This could be because the typical seasonal pattern of high winter SC and low summer SC was less pronounced during droughts compared to reference years (SI Fig. S.1C). Basin-wide, low SC (5th percentile) values increased by almost 44%, while peak SC (95th percentile) slightly declined by about 2% (Table 1). Salinity patterns were generally consistent across drought events, although fewer sites had data for 2001–02 ($n = 5$) and 2012 ($n = 14$). The low data availability was spread across sites with salinity declines and increases (SI Fig. S.6). Similar to WT, SC data were only available for non-reference sites in the central portion of the basin, with no coverage in the northernmost areas.

A sensitivity analysis testing different pre- and post-event baselines showed that the results are generally consistent with using the 3-year reference period (Table S.4). The notable differences were relative changes that were moderately lower using a 2-year baseline for AT (-22% instead of -25%), Q (-9% instead of -15%), and SC (3% instead of 5% when comparing the same sites). Using all normal years as a reference resulted in higher relative change for air temperature (31% instead of 25%). The all

normal years reference was not considered for SC and WT due to the low number of sites with data availability.

Drought recovery and trends from 1998–2022

Sites that had flow increases during drought returned quickly to baseline levels within 1–3 years (SI Fig. S.9). Most sites that experienced flow declines during drought recovered to their pre-event baselines within 10 years, with a median recovery period of 3 years (median recovery for 2001–02 drought = 4 years, for 2012 drought = 2 years; SI Fig. S.9). For the sites where flows declined, a small fraction (25 out of 202; SI Fig. S.9, SI Tables S.5 and S.6) had not recovered to pre-drought baselines after the droughts in 2001–2002 ($n = 20$) and 2012 ($n = 6$; 1 site did not recover from both 2001–02 and 2012 droughts). However, after the 2018 drought, a majority of the sites ($n = 117$) had not returned to pre-drought baselines as of 2022. The patterns for runoff efficiency (RE) were different from Q (Table S.5), with a few sites not recovering after 2001–2002 ($n = 3$), and more sites not recovering after the 2012 ($n = 20$) and 2018 ($n = 35$) droughts (Table S.6). In general, reservoir storage declines during drought are noticeable for all drought periods (SI Table S.5), with the larger reservoirs (Lake Powell, Flaming Gorge, Navajo, Strawberry) having low levels for a few years after the droughts ended (SI Fig. S.10).

Most sites with WT changes recovered quickly within 1–2 years, though a few took longer to recover. For SC, most sites recovered within 4 years of the 2001–02 and 2012 droughts. However, the 2018 drought had more lasting impacts, with 12 sites that had not returned to pre-drought levels as of 2022.

The Mann-Kendall tests showed that most sites did not have a significant trend between 1998 and 2022 for Q ($n = 184$) and RE ($n = 182$), with flows returning to near pre-drought levels after each event (SI Fig. S.11). Fifteen sites had significant decreasing trends for Q and RE, while a few sites had an increasing trend ($n = 3$ for Q, $n = 5$ for RE). Most sites also did not have a significant trend in WT (32 out of 36) or SC (19 out of 23). Four sites had significantly increasing trends in WT. Two sites had a significant increasing and decreasing trend in SC, respectively. Of the 43 reservoirs in the basin, 29 did not have a significant trend in storage levels, 12 had a decreasing trend and 2 had an increasing trend. Generally, the difference between the reference periods of the 2012, 2018, and 2021–2022 droughts relative to the 2001–02 drought was typically within 0–15%, with the exception of AT during the 2018 drought (Table S.7). Together, these results suggest that there were no long-term baseline shifts during our period of analysis for a majority of the sites across the basin.

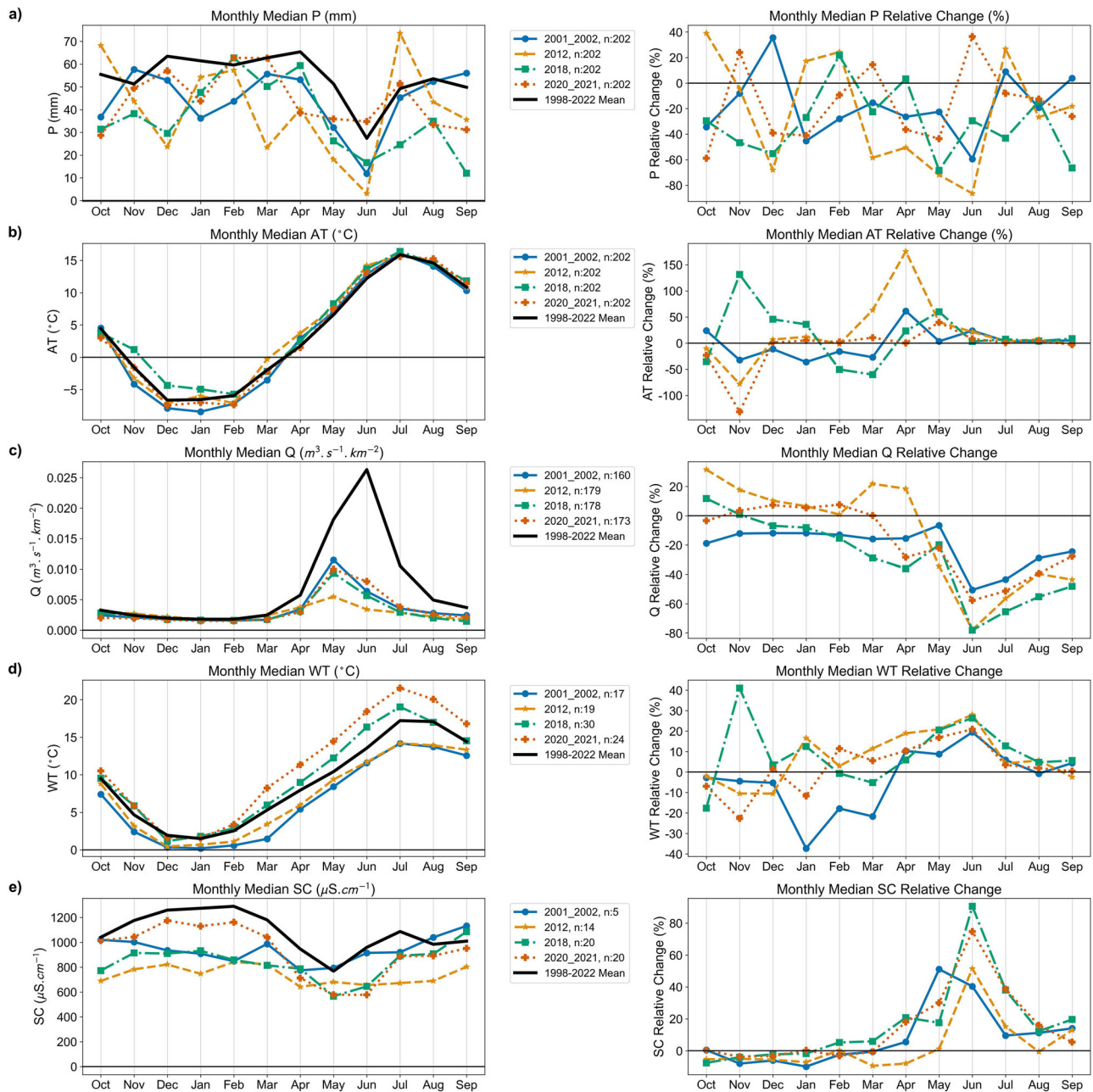


Fig. 2 | Monthly median (left panel) and relative change values (right panel). **a** precipitation, **b** air temperature, **c** streamflow (Q), **d** water temperature (WT), and **e** salinity (SC), separated by drought episode. The X-axis represents months ordered

by water year. The number of sites for which data were available for each variable and drought episode is shown as 'n' in the legend. Relative changes shown in the right panel are median values across all sites, calculated using Equation (2).

We compared annual precipitation-runoff (P-R) relationships during drought and reference years (Methods Section 5.6) to further evaluate drought impacts and recovery. Most sites (198 out of 202) did not have statistically significant changes in precipitation-runoff relationships during drought versus non-drought years. Three sites had positive shifts, where reductions in runoff (R) during drought were smaller than expected for a given precipitation decrease, with an average shift of approximately 0.66. One site experienced a negative shift, with greater reductions in R during drought than expected for a given precipitation decrease (SI Fig. S.12).

Catchment attributes influencing drought response

Grassland and deciduous forest cover, freshwater withdrawals, summer precipitation, and elevation showed moderate to weak negative correlations

(ρ between -0.2 to -0.4) with relative changes in Q (SI Fig. S.13). In contrast, the extent of shrubs in the watershed and riparian zones, topographic wetness index and baseflow index had moderate to weak positive correlations (ρ between 0.2 to 0.3) with relative changes in Q. The attribute z-score analysis indicates that the 32 catchments with increases in Q during drought are characterized by higher shrub cover, topographic wetness index, available soil water capacity, and finer soil texture indicating the potential for increased runoff (SI Fig. S.14). These catchments also had lower precipitation, grassland and forest cover, bulk soil density, coarser soils, slope, and elevation. Most of these sites were in the Western part of the basin, and had lower median flows as noted above (Section 2.2). Results for catchments with negative relative changes are not presented. Since these catchments comprise 84% of the dataset, their z-scores are substantially lower and less informative. Thus, the z-score analysis is only used to

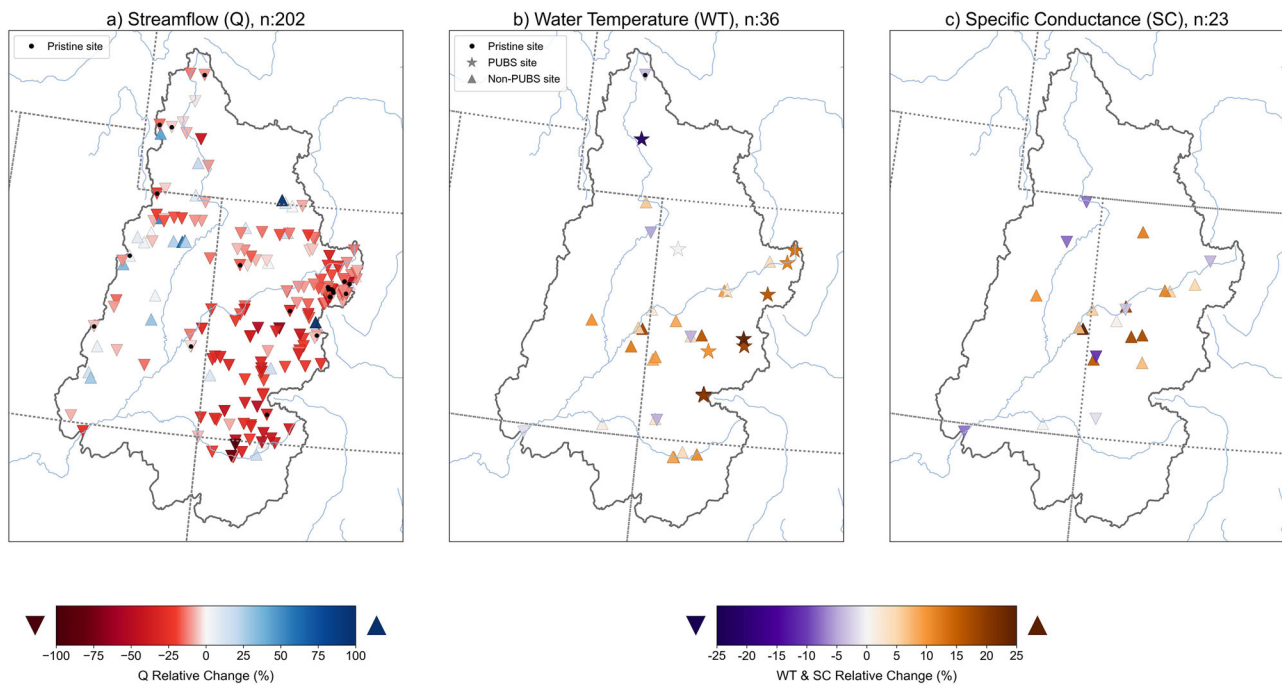


Fig. 3 | Annual median relative change (%) across all four drought episodes. **a** streamflow (Q), **b** water temperature (WT), and **c** salinity (SC). Each triangle reflects the relative change for that site during drought compared to reference periods, averaged for episodes for which data were available. The color shading indicates the magnitude of change and the direction of the triangle indicates positive (upward) or negative (downward) change. Star shaped markers for WT (12 sites)

indicate that the data for the site were estimated using a deep learning long short-term memory (LSTM) model for predictions in unmonitored basins (PUBS). The black dots indicate pristine sites (24 for Q, 1 for WT). Note the differences in color scales between Q and water quality (WT, SC). The UCRB shapefile boundary is obtained from the USGS¹²⁹. Elevation across the basin is shown in SI Fig. S.24 with data obtained from the USGS¹³⁰.

Table 1 | Median, peak, and low values of Q, WT, and SC

		Q ($\text{m}^3 \cdot \text{s}^{-1} \cdot \text{km}^{-2}$)	WT (°C)	SC ($\mu\text{S} \cdot \text{cm}^{-1}$)
Median (50th Percentile)	Drought ^a	0.0021	9.8	961
	Reference	0.0026	9.1	853
	Relative Change (%)	− 14.6	8.5	5.4
Peak (95th Percentile)	Drought	0.012	19.7	1304
	Reference	0.025	18.7	1260
	Relative Change (%)	− 47.3	4.7	− 1.7
Low (5th Percentile)	Drought	0.0011	0.77	541
	Reference	0.0013	0.54	385
	Relative Change (%)	− 14.7	2.0	44.0

^aDrought and reference year values are calculated as per site averages across reference or drought years, then the median value of all sites. Relative change is also the median value of all sites, averaged across all drought episodes.

interpret anomalous behavior during droughts such as increased Q, and decreased WT and SC.

Given the limitations with the results from the Spearman correlation and attribute z-score analyses, we used an XGBoost model with recursive feature elimination and two importance scoring methods to identify catchment attributes influencing streamflows, after excluding outlier catchments (Methods, Section 5.6). Twelve attributes were identified using the Shapley Values (RMSE=0.72; average test R^2 =0.42; average training R^2 =0.96). In contrast, twenty attributes were identified using the permutation feature importance score (RMSE = 0.72; average test R^2 =0.42; average training R^2 =0.98). The considerable difference in performance for training and test sets suggests possible overfitting, despite the extensive hyperparameter tuning, possibly due to the small size of the target dataset. However, the top attributes selected were mostly consistent across numerous ensemble runs, model configurations (with and without outliers), and scoring methods, providing confidence in the results. Feature rankings varied slightly across the different model runs.

From the XGBoost model, factors associated with lower streamflow declines during drought were primarily higher baseflow index and soil permeability with slight differences between the Shapley values (Fig. 4A) and permutation feature importance (SI Fig. S.15 A). Lesser influences were greater mean aspect northness, June precipitation, and average sand content. Greater declines were associated with higher freshwater withdrawal rates, deciduous forest cover within the riparian zone and 100 m of main stem, average silt and clay content, subsurface flow contact times (residence time of infiltrated water in the saturated subsurface zone before discharging into the stream), and mean annual air temperatures. August precipitation and features associated with anthropogenic development were also important, but had weak or mixed directionality. The top three attribute categories were consistent between the Shapley and permutation feature importance scores (Fig. 4B and SI Fig. S.15 B). Attribute categories associated with lower flow declines during drought include Baseflow Index and Aspect and, to a lesser degree, Soil Texture and Water Holding Capacity (Fig. 4B, SI Fig. S.15 and S.16). Conversely, Deciduous Forests and Freshwater Use were associated with decreased flows during drought, suggesting these factors contributed to higher catchment vulnerability. Across the UCRB, baseflow index ranged from 50–75% (SI Figs. S.16 and S.17, SI Table S.8), with the lowest values in the southern catchments. Deciduous forest cover ranges from 0–55%, with most catchments having less than 20%.

These results reinforce the findings from the correlation and z-score analyses, with the XGBoost model capturing more complex, non-linear relationships between flows and catchment attributes. For these analyses, we identified both individual attributes and aggregated importance scores using a network clustering approach (see Methods Section 5.6) because many catchment attributes are redundant (e.g., percent deciduous forest within 100m mainstem, deciduous forest within 100m riparian). Our interpretation focuses on the directionality and aggregate importance of the attributes, rather than their individual rank order.

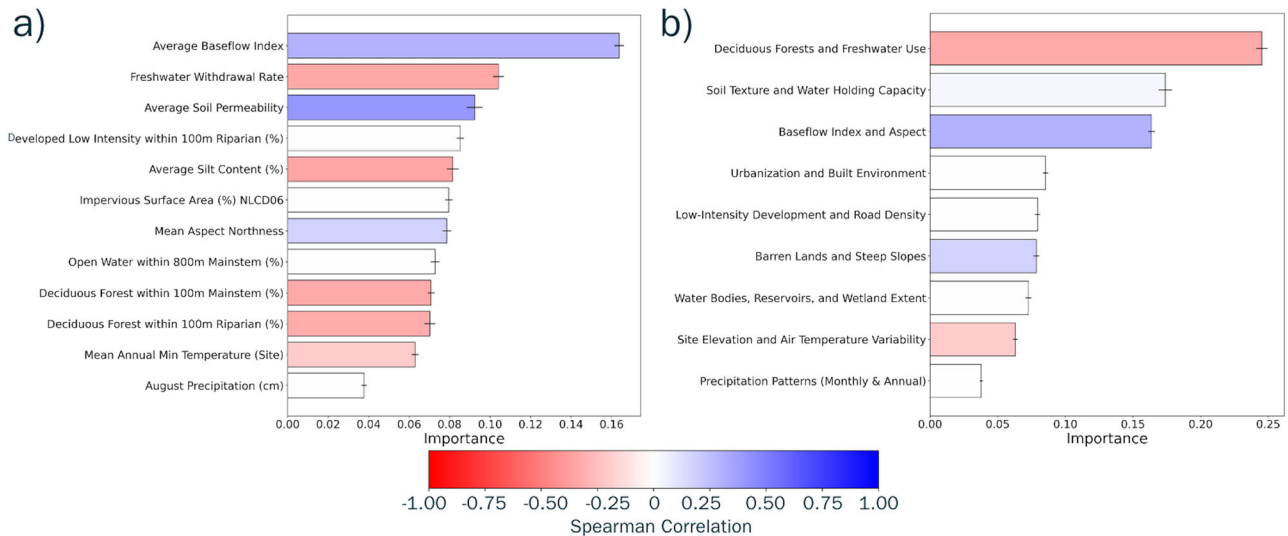


Fig. 4 | Shapley feature importance values identified using recursive feature elimination with XGBoost models. a Attributes (renamed from the GAGES-II dataset for conciseness and clarity) and **b** attribute categories. Feature importance values are normalized to sum to 1. Bar colors indicate the Spearman correlation

between the GAGES-II attribute values and the feature importance. A positive correlation (blue) indicate attributes associated with lower declines in median flows during drought, while negative correlations (red) indicate greater drought impacts.

Relative changes in WT are positively correlated with precipitation, the extent of woody wetlands and forests in the riparian zone (within 800 m of mainstem), barren lands, steep slopes and high elevations (ρ approximately 0.6; SI Fig. S.18). They also had strong negative correlations with agricultural land use as indicated by nitrogen and phosphorus applications (ρ approximately -0.8) and other related variables such as pesticide use, and cultivated land (ρ approximately -0.6) as well as the extent of shrub cover (ρ = -0.6), subsurface contact time index ρ approximately -0.6), and finer soil textures and available water capacity (ρ for the category is approximately -0.35). The 5 catchments with negative changes in WT are characterized by lower soil infiltration rates, higher shrub and cultivated land cover, and lower extent of minimally protected areas (e.g., BLM lands, wildlife management areas, and game and fish preserves), woody wetlands, and lower erosivity (SI Fig. S.14). To evaluate whether site locations could be a factor influencing observed correlations, we calculated the attribute z-scores of the WT sites relative to all 202 sites in the UCRB (Section 5.6). The attributes of the WT catchments are within one standard deviation of all sites, indicating that the WT sites are broadly representative of the 202 sites included in this study.

Relative changes in SC have moderate negative correlations with water bodies and storage capacity (ρ between -0.5 to -0.7), and a moderate positive correlation with developed open spaces within 100 m of mainstem rivers (ρ approximately 0.5; SI Fig. S.19). The 6 catchments with negative relative SC changes are characterized by the greater presence of water bodies, wetlands, and reservoirs as well as lower northern aspect, elevation, slope, air temperature, and precipitation (SI Fig. S.14).

The XGBoost model could not be used for WT and SC due to the small number of catchments with available data. We note that the limited sample size may also bias the results from the z-score and correlation analyses.

Discussion

Severe meteorological drought episodes between 1998 and 2022 in the UCRB clearly caused negative impacts on water quantity and quality. Most sites experienced annual decreases in streamflow and increases in water temperature and salinity during drought years compared to adjacent years with normal conditions. The greatest impacts occurred during the spring and summer months, while the fall and winter months were less affected. Although meteorological drought intensities, quantified using the SPEI (Fig. 1), were relatively uniform basin-wide, catchment responses varied

substantially due to differences in their biophysical and anthropogenic attributes.

Across all episodes, we find that average flows (normalized by basin area) declined by 15% during drought compared to recent non-drought years using a 3-year pre- and post-drought reference period (see Section 2.2 for a sensitivity analysis with different baselines). Although the onset of meteorological drought was driven by precipitation deficits and elevated pre-drought air temperatures in 2012 and 2018 (Figs. 2 and S.11), precipitation-runoff relationships did not change significantly at most sites for drought years relative to normal years (Section 2.3). The absence of shifts in precipitation-runoff relationships, combined with the lack of trends in Q and RE, as well as their post-drought recoveries to pre-event baselines (Fig. S.9 and SI Table S.5), implies that individual meteorological drought episodes did not have a persistent effect on streamflows in the UCRB from 2000 to 2018 for most sites ($n = 177$, with 25 sites not recovering from earlier droughts). The 2018 episode had the greatest reduction in spring and summer flows (Figs. 2 and S.1), and most sites ($n = 118$) had not recovered by 2022. This persistence occurred despite post-drought precipitation exceeding pre-drought baselines, and post-drought air temperatures being lower than pre-drought baselines, unlike for other episodes (SI Table S.5). The sustained impact of the 2018 drought on streamflows is likely because the 2020–21 drought followed immediately after, leaving insufficient time to recover given median recovery times of about 3–4 years. In contrast, streamflows after the 2012 drought recovered quickly (mostly within 2 years, Fig. S.9), likely due to abnormally wet conditions at many sites in 2011 and 2014 (Fig. 1). The 2001–02 event had the most intense drought year during the period (SI Table S.1), and had slower recoveries (Fig. S.9), with substantial differences between the pre- and post-drought baselines in Q (– 30% for 147 sites with no significant or increasing trends) and RE (– 16% for 147 sites with no significant trends). In contrast, in 2020–21, there were smaller differences between pre- and post-event baselines for Q (– 18% for 159 sites with no significant trends), but much larger differences in RE (– 36% for 158 sites). Although lower RE can occur in years with lower precipitation or warmer spring temperatures⁶¹, the similarities in the P and AT patterns in 2001–02 and 2020–21 (Fig. 2) indicate that this may not fully explain the differences between the 2001–02 and 2020–21 events, and that the sequencing of drought events can have an effect on RE. The few sites with decreasing trends had major differences between pre- and post-event baselines for both Q (–25 to 52%) and RE (6 to – 49%) for most of the

drought episodes. Reservoir releases buffered flows during all droughts. In particular, the difference between pre- and post-drought baselines for reservoir storage in 2001–02 was 32%. Pre-drought reservoir storage in 2018 was only about 4% lower than in 2001–02, whereas pre-drought reservoir levels in 2020 was 15% lower relative to 2001–02.

There was substantial spatial variability in streamflow changes during drought across different catchments, despite relatively uniform changes in climatic forcings (i.e., meteorological drought intensities). Catchments with higher freshwater withdrawals, deciduous forest cover, mean annual air temperatures, finer soils (silt or clay), and longer bedrock residence times (Fig. 4, SI Figs. S.15 and S.16) were more susceptible to streamflow declines during drought. Factors leading to streamflow resilience are subsurface flows, higher soil water retention and infiltration capacity (e.g., soil permeability, sand content), and to a lesser extent northern aspect and June precipitation. Increases in streamflow during drought were noted in about 15% of catchments, mainly in the Western UCRB, that had predominantly low flows, lower elevations, arid climates, higher shrub cover and topographic wetness index, finer soil texture, and lower grassland and forest cover.

Our findings of basin-wide declines in flows during drought, particularly during the summer, is consistent with studies showing decreasing peak flows from long-term trend analysis or model projections under future climate scenarios^{62,63}. However, the absence of trends in flows and runoff efficiency between 1998 and 2022 contrasts with other studies that generally find declining flows over longer periods, particularly when comparing the recent drought as a whole to pre-2000 conditions^{18,45,62}. Streamflows in the basin are driven by precipitation, and mostly snowmelt. The variability and trends in precipitation in recent years contributed to approximately half of reduced streamflows in the basin⁴¹. Prior studies show that drought in the region is increasingly driven by rising air temperatures^{18,64}, which can reduce streamflow with changes to snowmelt rates and timing, and higher evapotranspiration⁴⁵. In particular, air temperature and antecedent soil moisture conditions can amplify the effects of precipitation deficits in reducing streamflows during drought¹⁷. Changes in snowpack volumes and melt timing, driven by changes in P and AT during drought, also result in changes in peak flows and water temperatures with cold meltwater runoff to streams earlier in spring^{42,43,47–50,65}.

Our results show that while lower precipitation and elevated air temperatures are important in causing reduced streamflows during drought (Fig. 3), the directionality and magnitude of relative changes in Q are also determined by catchment properties, which control water storage, consumption, and release. This finding aligns with a recent study showing that climate only explains 54% of streamflow variability between 1986 and 2021, suggesting that anthropogenic activities or other watershed processes are also important⁶⁰. For example, the role of freshwater withdrawals (indicative of total human use across agriculture, industry, and domestic sectors) in amplifying flow declines during drought is expected, with substantial consumptive use from irrigation in the UCRB headwaters⁴⁵. In contrast, the importance of natural factors influencing flows during drought across many sites in a heavily managed basin like the UCRB is surprising. Prior studies in the UCRB have linked drought impacts on streamflows with vegetation and forest cover^{37,62}, as increased vegetation water use combined with decreased precipitation or recharge can reduce flows. Groundwater storage can buffer streamflow against immediate drought impacts in the basin, but forest cover and elevated summertime evapotranspiration can reduce this storage³². In particular, older, deeper groundwater in permeable bedrock is more effective at sustaining minimum flows in dry years⁶⁶, but this comes at a cost if recharge in subsequent wet periods is insufficient to enable recovery. A recent analysis of satellite observations⁶⁷ found that the UCRB lost 14.6 ± 3.5 km³ of terrestrial water storage between April 2002 and October 2024, of which 53% was estimated to be from groundwater (7.8 ± 5.3 km³), 36% from reservoirs (5.3 ± 0.8 km³), and the remaining loss from soil moisture storage (3.8 ± 4.0 km³). In contrast, the snow water equivalent increased by 2.3 ± 0.3 km³ during this period. In particular, groundwater declines were higher and more persistent for several years in the periods following drought episodes

(see Fig. 2 in ref.⁶⁷) compared to reservoir storage. A recent study using tritium age dating showed that groundwater is not only important for baseflow, but also dominates runoff during snowmelt⁶⁸. Our analysis is consistent with these previous studies, and also finds that reservoirs and groundwater buffer streamflow during droughts, where sites with lower infiltration capacities and baseflows have reduced capacity to withstand droughts. The lack of significant shifts in precipitation-runoff relationships during drought is also consistent with findings from other drought studies for catchments with similar biophysical attributes^{13,21,69}. Notably, catchments dominated by subsurface flow and water storage had smaller shifts in precipitation-runoff relationships during drought than surface-dominated runoff^{13,69}. Drier, flatter, and less forested catchments were more likely to exhibit changes in precipitation-runoff relationships when investigated in Australia²¹. Taken together, these suggest the important role of groundwater storage and recovery in mediating both immediate and long-term streamflow responses to drought in the UCRB.

Water temperature and specific conductance increased during drought by 9% and 5% across 36 and 23 catchments, respectively, using the 3-year pre- and post-drought reference period. This is consistent with observations of increases in summertime WT and salinity at a few sites in summer 2002^{58,59}. The impacts on WT were transient, with most catchments returning to baselines within 1 to 2 years. The rapid recovery of WT is also reflected in the lack of significant trends at most sites. Apparent WT increases in 2018 and 2020–2021 relative to 2012 and 2001–02 are partly explained by unusually warm temperatures in November 2018, and partly because the 2021 data were from warmer sites than for earlier periods. WT changes varied across the episodes, with a consistent pattern of increases in spring and summer (Figs. 2 and S.8). Like Q, post-drought SC returned to pre-drought baselines until 2018 with recovery periods ranging from 1 to 6 years. The successive droughts since 2018 had longer lasting impacts on SC, although most sites did not have significant trends. A surprising finding is that despite increases in relative change of SC in summer months, monthly median SC was lower than average across the basin in all drought episodes (Fig. 2E and SI Figs. S.2 and S.8). This pattern is not explained by low data availability of SC in earlier droughts (SI Fig. S.6). A possible explanation is that slight declines in peak SC in the winter are not offset by increases in low SC in summer during drought. Data were limited to analyze the cascading impacts of Q on WT and SC, with only 9 sites having observations for all three response variables for at least one drought episode (SI Fig. S.20). In general, simultaneous declines in Q and increases in WT and SC occurred mainly in the spring and summer (Fig. S.21).

Similar to Q, there was large spatial variability in WT and SC responses to drought. While WT changes are driven by AT, catchment attributes also appear to have an influence on drought response. WT relative changes are positively correlated with slope and elevation, precipitation, riparian woody wetlands and forests, and negatively correlated with agricultural and shrub lands, subsurface residence times, and finer soil textures (Fig. S.18). This implies that high-elevation watersheds with lower baseline WT, more snowpack and riparian vegetation see greater increases in WT during drought, while lower-elevation watersheds with higher baseline WT, shrubland and agricultural areas are less affected. This suggests that WT changes may additionally be driven by snowpack volumes, melt timing and runoff, shading, and agricultural return flows. Catchments with open water bodies and reservoirs were more resilient to increases in SC, possibly because reservoirs can act as sinks for dissolved solids in the UCRB⁷⁰. While most stations exhibit increasing WT and SC during drought, some show decreasing relative changes, such as the gaging station on the Paria River near Lees Ferry (station ID 09382000) located at the southwestern end of the basin. The z-score analysis enables the identification of catchment attributes that could potentially result in such anomalous behavior. The Paria River catchment is characterized by above-average temperatures, a greater proportion of barren land and higher-order streams, as well as below-average elevation, precipitation, and vegetation, suggesting this is an arid catchment (SI Fig. S.22). We note that our interpretation for WT and SC is limited by sparsity of data, which are almost an order of magnitude lower than for Q,

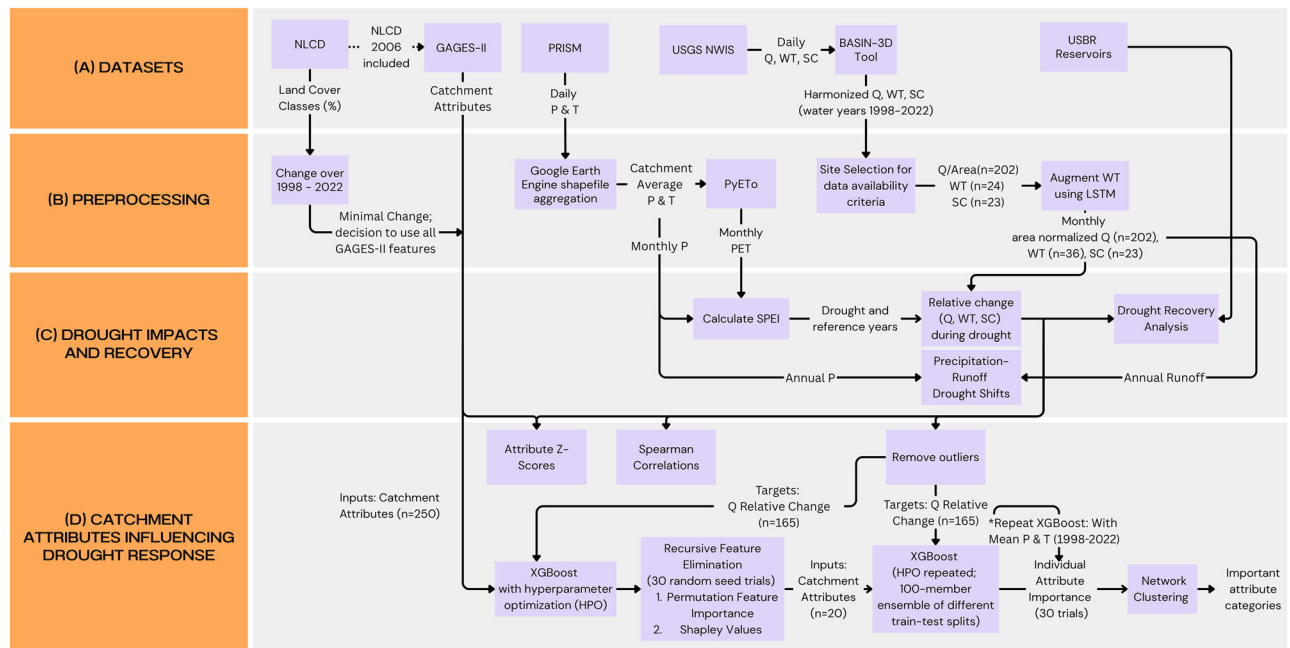


Fig. 5 | The DIRE (Drought Impacts and Resilience Evaluation) data-driven framework used in this study to identify catchment response to drought. Its modular and flexible design enables application to any variable of interest, using methods and metrics of choice to evaluate drought impacts and resilience.

with uneven spatial and temporal coverage across the basin for all drought episodes. For example, the lack of reservoir effects on WT may be an artifact because many of the sites sampled were not immediately downstream of a reservoir (SI Fig. S.23). We marginally improved WT coverage using the LSTM model, which has uncertainties (upto 2 °C) particularly for sites influenced by groundwater and reservoirs⁷¹. We did not have similar gap-filling models for SC. Using the event-based metric and the maximum number of sites available for each drought episode increased spatial coverage, but limited our ability to compare sites across droughts and determine their response and recovery. Additionally, the analysis is limited by the attributes present in the GAGES-II dataset, and other factors that are not present in the dataset, such as management decisions, thermoelectric plants, or deeper subsurface properties could affect catchment drought response.

In this study, we introduce DIRE, a data-driven framework that combines ML with other computational approaches to assess drought impacts and recovery for different response variables. While prior drought studies have implemented parts of this analysis, DIRE brings these elements together into a single framework that is robust, modular and flexible, where components can be swapped out to use alternate metrics or methods. We also introduce methods to analyze factors influencing catchment drought response. The network analysis enables clustering of redundant catchment attributes and easier interpretation of high-dimensional data ($n = 247$ catchment attributes). Our XGBoost modeling approach incorporates extensive ensembles, recursive feature elimination, and attribute grouping via clustering to address both redundancy in catchment attributes and the stochasticity in model construction. This approach helps avoid misidentifying drivers of streamflow change using more simplistic ML or statistical modeling approaches. A z-score analysis enables comparison of properties across different catchment groups, including those exhibiting outlier behavior. Collectively, the DIRE framework provides a general scaffold to analyze the impacts and resilience to droughts or other short-term disturbances on catchment functions.

Conclusions

Our study addresses key knowledge gaps on drought impacts in the UCRB, and other mountainous regions. We find substantial basin-wide declines in streamflows, and increases in stream temperatures and specific conductance during four severe meteorological droughts in 2001–02, 2012, 2018 and

2020–21. To our knowledge, there are no prior studies that assess post-drought recovery for these variables in the UCRB. Streamflow and specific conductance recovered after drought episodes through 2018, with recovery times ranging from 1 to 10 years and 1 to 6 years, respectively. However, successive dry years since 2018 have resulted in prolonged impacts on Q and SC. Water temperatures recovered more rapidly, typically within 1–2 years for all drought episodes. We show that in addition to climatic drivers, natural and anthropogenic factors strongly influence catchment responses at basin scale during short-term droughts. Notably, even when meteorological drought intensities were similar across the basin, catchment response varied in space and time due to their biophysical and anthropogenic attributes. After accounting for redundancy, factors associated with streamflow declines during drought include freshwater withdrawals, greater deciduous forest cover, higher mean annual temperatures, finer soils, and longer subsurface residence times. Factors driving streamflow resilience include subsurface flows and properties that enhance infiltration, and to a lesser extent northern aspect and June precipitation. Catchments with higher slopes, elevation, precipitation, riparian woody wetlands and forests had greater increases in WT. In contrast, higher shrub and agricultural land cover and lower infiltration were associated with smaller WT increases (greater resilience). The presence of open water bodies and reservoirs mitigate SC increases during drought. The DIRE framework provides a robust, modular approach to assess drought impacts, recovery and resilience that can be extended to other variables and disturbances.

Methods

For this study, we developed the DIRE (Drought Impacts and Resilience Evaluation) data-driven framework (Fig. 5), which is described further in the sections below. Data associated with this study are available publicly⁷².

Study site: Upper Colorado River Basin (UCRB)

The UCRB contains the headwater catchments of the Colorado River in Colorado, Wyoming, Utah and New Mexico, flowing into the Lower Basin at Lees Ferry in Arizona. Elevations in the UCRB range from 1200 to 4300 m (Fig. S.24) across low-lying, arid or semi-arid catchments to high-elevation alpine environments⁶³. The land cover is dominated by rangeland, shrubland, and evergreen forests^{63,73,74}, with few agricultural lands (2–3%)⁷⁰. Both temperature and precipitation follow a strong seasonal pattern and

elevation gradient, with cool season snow concentrated in the winter months followed by warming and snow melt in the spring, and intensifying heat in the summer³⁷. While spring snow melt from headwater catchments can account for up to 70% of runoff in the basin, approximately half of the streamflow is estimated to come from baseflow^{75,76}. Both seasonal and spatiotemporal differences in streamflows have been identified across both the Upper and Lower basins, with greater declines occurring during peak flows in the summer and at lower elevations (< 2300 m)⁶². There are 21 main reservoirs located in the UCRB⁷⁷ (43 total⁷⁸), which influence the timing of discharge as well as water quality⁵². The 1922 Colorado River Compact and other documents collectively known as the Law of the River governs how water is apportioned across basins, and expires in 2026. The Compact over-allocated water relative to actual average flows^{79–81}, and is therefore being updated to have modified operations that account for the drought affecting the basin⁸². The stricter allocations being considered combined with potential hydrologic shifts that may occur during future drought or arid conditions is expected to have noticeable downstream impacts on water availability and quality in many sectors^{80,81}.

Several studies have analyzed droughts and streamflow reductions in the basin over the past century and beyond^{17,62}. The first two decades of the century (2000–2021) were identified to be the driest period in the region since 800 CE⁴⁰. In contrast, other paleoclimate reconstructions found that this current drought was less severe than a megadrought in the second century^{83,84}. Increasing air temperature in the UCRB since the 1980s led to a 7% decrease in mean water-year streamflow⁴². At Lees Ferry, the monitoring station considered representative of the conditions in the UCRB, average flows decreased by 17% from 18.7 billion cubic meters (15.2 million acre feet) in 1906–1997 to 15.5 billion cubic meters (12.6 million acre-feet) in 1998–2022 (calculated from Reclamation, 2024⁸⁵). About half of this long-term decline in flows has been attributed to climate change⁸⁶, with about a one-third reduction occurring due to warming temperatures alone. In addition, streamflow drought duration and deficit increased from 1950 to 2020 in the region^{3,49}.

In contrast to the vast amount of literature on flows in the basin, there are only a handful of studies documenting water quality trends and drought impacts in the UCRB. Salinity (measured as specific conductance or total dissolved solids) originates from both natural (e.g., bedrock) and anthropogenic (e.g., agricultural runoff) sources⁴¹. Most (82–89%) of the dissolved solids in streams that contribute to salinity originate from ground water inputs^{87,88}. Salinity has been a historical problem in the basin, costing approximately \$400 million of economic damages annually, and is regulated under a US-Mexico agreement^{41,89}. Rigorous monitoring and salinity control programs were implemented in the 1980s to address economic and safety concerns under the Clean Water Act, resulting in substantial declines in salinity between 1980 and 2000^{41,70}. Factors influencing stream temperatures include air temperature, runoff from snowmelt and groundwater⁵⁷. Dams and reservoirs are also used to regulate water temperature^{90,91}, and can influence dissolved-solid retention⁷⁰. However, at times, reservoir storage can have unintended consequences. For example, when flow regulation began by filling Lake Powell in 1980, average water temperatures decreased beyond the Glen Canyon Dam resulting in considerable impacts on aquatic life, requiring changes to dam operations to mitigate its effects^{92,93}.

Data acquisition

This study focuses on water quantity and quality variables: streamflow (Q ; $m^3 s^{-1}$ normalized by catchment area to $m^3 s^{-1} km^{-2}$), water temperature (WT, °C), and specific conductance (SC, $\mu S cm^{-1}$ at 25 °C) as a proxy measure of salinity. Data for these variables for water years 1998 to 2022 were compiled from the USGS National Water Information System (NWIS) database for the UCRB (HUC Code 14)⁹⁴. Data were obtained using BASIN-3D (Broker for Assimilation, Synthesis and Integration of Environmental Diverse, Distributed Datasets), a brokering tool designed to integrate time series data from diverse data sources into standardized formats⁹⁵. Daily mean values for the variables were used when available, and averages of

minimum and maximum values were used for sites that did not have a reported mean. Streamflow values of -99999, negative flows, and those associated with 'ICE' were dropped and marked as *NaN*.

Meteorological data for water years 1998–2022 were compiled from the PRISM (Parameter-elevation Regressions on Independent Slopes Model)⁹⁶ dataset at 4 km resolution through the Google Earth Engine (GEE)⁹⁷ platform. For each catchment with streamflow and catchment characteristic data, precipitation and air temperature values were aggregated as monthly, catchment-wide averages using the Mean Reducer function in GEE, by applying the corresponding HUC-8 catchment boundary shape files from GAGES-II.

Data for 250 catchment attributes were obtained from the USGS GAGES-II (Geospatial Attributes of Gages for Evaluating Streamflow)⁹⁸. The dataset contains over 350 physical attributes for over 9300 catchments (HUC8 size) in the United States that either had more than 20 complete years of streamflow data since 1950 or was actively monitored as of water year 2009. We identified pristine and human-impacted sites using the reference and non-reference classification from the GAGES-II dataset. Storage level data for 43 reservoirs were downloaded from the US Bureau of reclamation website⁷⁸. Additional land cover characteristics were compiled through extracting land cover percentages for counties in the UCRB using the Multi-Resolution Land Characteristics (MRLC) Consortium Web Viewer tool⁹⁹.

Pre-processing

We applied the following data availability criteria to adequately capture the temporal evolution of drought, while accounting for the importance of seasonality and snow melt: (1) A valid year of data should have at least 11 valid months of observations, and (2) each month should have at least 10 days of measurements for at least one variable. Once drought and reference years were identified (Section 5.4), only sites with valid data for the entire drought episode were used. Additionally, sites with streamflow data were downselected based on the availability of corresponding catchment attributes in the GAGES-II dataset. Flows were normalized by catchment area (in square kilometers) obtained from GAGES-II. In contrast, all sites with WT and SC measurements were included regardless of whether corresponding information was available in GAGES-II, to maximize data availability. We chose to retain sites close to dams and reservoirs to capture the potential influence of these structures on drought response. Due to differences in gap lengths across sites and variables, we did not temporally gap fill records to maintain consistency and minimize uncertainties in this analysis.

After applying the above-mentioned criteria for site selection within the study period, there were 202, 24, and 23 sites with Q , WT, and SC measurements, respectively (Table S.3). Most sites for each of the three variables were human-impacted, but 24 sites for Q and one for WT were pristine. The average physical attributes of the selected sites that had associated GAGES-II data are summarized in SI Table S.8.

Only 24 sites with stream temperature observations met our data availability criterion, and are located in the central and Eastern parts of the basin. (Section 5.2). Hence, we used a deep learning model to estimate WT for 12 additional sites, which expanded our analysis to 35 sites (SI Table S.9). The continental-scale (CONUS) long short-term memory (LSTM) model is described in refs. 71, 100 and predicts WT time series in unmonitored sites using dynamic meteorological data (e.g., air temperature, solar radiation, day length), dynamic streamflows, and 274 static site attributes from the GAGES-II dataset as inputs. This set of site attributes consists of all non-categorical attributes available in GAGES-II. Though this work is focused on the UCRB, recent literature has revealed that large continental, top-down models are superior to region-specific and local alternatives for deep learning models like LSTM¹⁰¹, which informed our choice to use a continental-scale stream temperature model for this study. The LSTM architecture was chosen because it is the most common deep learning architecture for time series prediction of hydrologic variables in unmonitored sites¹⁰¹. For further details on the LSTM recurrent neural network

architecture, see ref. 102, and details of our implementation are presented in ref. 71. The total training data consisted of time series from 782 stream sites and over four million WT observations across the conterminous United States from 1980–2021⁷¹. The model has an estimated median per-site RMSE of 1.43 °C for spatiotemporal predictions at unmonitored sites in the CONUS. Only sites that had an RMSE of less than 2 °C in the UCRB were included (SI Table S.9).

We calculated daily runoff from daily streamflow discharge (Q) as (Equation (1)):

$$\text{dailyR(mm)} = Q(\text{m}^3\text{s}^{-1}\text{km}^{-2}) \times 86.4 \quad (1)$$

Then we calculated runoff efficiency (RE as runoff/P, dimensionless) by summing daily runoff and monthly precipitation (P) to annual water years (mm/year), and dividing annual runoff (R, mm) by annual precipitation.

A limitation of the GAGES-II dataset is that it uses the 2006 National Land Cover Dataset (NLCD) and 1970–2000 PRISM dataset for meteorology. We assessed the suitability of this dataset for our analysis by calculating the degree of change in land cover over the 24-year study period. For this, we compared differences in NLCD land cover classes between 1998 and 2022⁷⁴ from the extracted land cover percentages for counties in the UCRB. We subtracted land cover percentages between 1998 and 2022 to estimate net changes in land cover. The percentage of each county inside the UCRB was calculated using the Overlay Layers analysis tool within ArcGIS on the US Census 2020 county boundaries¹⁰³ and the Colorado River Basin boundaries¹⁰⁴. The land use and land cover in the basin was mostly the same over the study period, typically changing by < 5% and not more than 14% (SI Table S.10). Only six counties had land cover changes of > 5% in at least one class, although some of these did not contain substantial areas within the UCRB. Specifically, three counties in Colorado (Grand, Larimer, and Mineral with 99%, < 1%, and 25% area within the UCRB, respectively) had 11–14% decrease in evergreen forests and 12–14% increase in grasslands or shrublands. La Plata, CO, Utah, UT, and Teton, WY (100%, 5%, and < 1% area within the UCRB, respectively) had 5–9% increases in shrublands, while Teton, WY also had an almost 9% decrease in grasslands. The meteorological variables in GAGES-II were calculated from 800 m PRISM data over a 30 year period of record (1971–2000). This is considered representative of historical climate prior to the onset of the drought in our study. Since average meteorological conditions have shifted between 1970–2000 and 1998–2022^{37,39}, using more recent temperature and precipitation data could alter importance scores in the XGBoost model (Section 5.6). We performed three additional ensemble simulations in which the RFE-selected traits were augmented with climatic variables from 1998–2022 aggregated as an overall average, and at the seasonal and monthly scales. In all cases, the RMSE and R² of the test sets did not improve and remained within one standard deviation of the original simulation (SI Table S.11). This result is consistent with the feature importance analysis (Fig. 4 and SI S.15), which shows that climatic variables are not strong predictors for relative changes in streamflows. We therefore retained the original GAGES-II climatic variables from 1971–2000 as the basis for our analysis”.

Identification of meteorological drought

We used the Standardized Precipitation Evapotranspiration Index (SPEI) to identify meteorological droughts, which is widely used to incorporate the effects of both precipitation deficit and temperature increases^{105,106}. The SPEI was calculated for all sites that had meteorological data by subtracting the potential evapotranspiration (PET) from precipitation (P) as a measure of the water surplus or deficit (P–PET). We used Thornthwaite’s equation¹⁰⁷ to calculate monthly PET using the PyET Python package¹⁰⁸, using the variables present in PRISM. While other methods to calculate PET have been found to be more accurate for arid and semi-arid regions¹⁰⁹, Abatan et al. (2017)¹¹⁰ found no significant difference in mean SPEI in the UCRB using the Thornthwaite method compared with the Hargreaves¹¹¹ and Penman-Monteith¹¹² methods for recent drought episodes since the 2000s.

We calculated the average (P–PET) at each site across a window of 12 months, ranked the values, and calculated the anomalies based on the 1998–2022 average using the SPEI Python package¹¹³. We used the Gringorten Plotting Position¹¹⁴, which is a non-parametric empirical plotting approach to transform the data to a standard normal distribution and used by many studies^{115–117}, instead of assuming a probability distribution (e.g., gamma or log-logistic distributions).

We then classified drought and non-drought years using the annual mean SPEI values averaged across the basin. The United States Drought Monitor (USDM)^{118,119} considers SPEI values of -0.49 and above as “Normal or wet conditions”, -0.5 to -0.79 as “D0—Abnormally Dry”, and anything below -0.8 as further levels of drought (moderate, severe, extreme, and exceptional). Abatan et al. (2017)¹¹⁰ classified SPEI scores in the UCRB between 1 and -1 as “Normal”, with negative values beyond -1 indicating drought severity. Following this, we chose to classify drought periods as years with SPEI values less than -1 , and “Normal” years as those with SPEI values between 1 and -1 . Normal years adjacent to a drought are hereafter referred to as “reference periods” or “reference years”. Reference years with SPEI between -1 to 1 were identified from the three years prior to or following each drought event (SI Table S.1). We verified that using some alternate SPEI values as thresholds for defining drought and reference periods (e.g. drought SPEI less than -1 and reference values between -0.5 to 0.5) did not affect the results in this study.

Drought impacts and recovery for streamflow, specific conductance, and water temperature

Drought impacts have often been determined based on comparisons to long-term baseline conditions, such as the last two to three decades preceding drought^{58,59} or over the last century^{18,120}. For example, standardized streamflow indices have been used to quantify streamflow deficits and drought duration in the UCRB from 1921 to 2020³. Another approach for determining the effects of a particular drought episode is based on identification of short-term changes relative to a pre- and post-event baseline. We followed the latter by adapting methods outlined in van Vliet and Zwolsman (2008)²⁴ to calculate the impact of particular drought events on Q, WT, and SC using event-based metrics. We chose this approach to avoid incorporating long-term trends in the variables and to decouple the impacts of drought from the effects of warming¹²¹ and human interventions (e.g., salinity control program). Once severe drought years were identified using the SPEI (Section 5.4), reference years were identified as neighboring years up to three years before and after drought episodes that had “Normal” SPEI scores.

For this analysis, monthly, seasonal and annual median Q (normalized by basin area), WT, and SC were calculated for each year. Median values were chosen instead of average or cumulative values to avoid skew from outliers, inconsistencies due to gaps in the data, and for comparison across the three different variables. For each drought episode, we used the median of pre- and post-event reference years together to represent baseline values during the reference period. Thus, reference conditions were based on a minimum of at least one “Normal” SPEI year to a maximum of six years. For drought episodes that were more than one year long, we used the median of the drought years. Seasons were defined as in Carroll et al. (2024)³² based on the water year and peak snow accumulation in the basin typically occurring on April 1. The seasons are fall (October–December), winter (January–March), spring (April–June), and summer (July–September).

Drought impacts were calculated as the average relative change for a variable during a drought episode compared to the baseline (Equation (2), where variable X represents Q, WT, or SC):

$$\Delta X = \left(\frac{\text{DROUGHT_X} - \text{REFERENCE_X}}{\text{abs(REFERENCE_X)}} \right) * 100 \quad (2)$$

We then calculated the median value of the relative changes for each variable across all sites to investigate the basin-wide impact per episode, again to avoid bias from outlier sites. Each drought episode had a different

set of sites for the 3 variables, since we chose all sites that met the selection criteria (SI Table S.3). This maximized data availability by allowing use of the greatest number of sites that had data for each episode without requiring observations for the entire 40-year period at each site and for each variable. We then calculated the aggregated impact across all events by averaging the relative change across all drought episodes for each site. Depending on data availability, this was a single number for sites that only had data for one event, or an average of two to four events for other sites (SI S.6). In addition, we calculated the annual average 95th and 5th percentile for each variable, again taking the median of these values across years and sites to explore changes in minimum and maximum values.

We conducted a sensitivity analysis to test the effects of different pre- and post-event baselines on the results. Specifically, we calculated relative changes using (1) a 2-year reference period (instead of the 3-years used in this study), (2) a 3-year reference period only considering the years prior to the drought, and (3) the average of all normal years across the period of study.

To examine catchment recovery after drought, we first investigated trends for P, AT, Q, RE, WT, SC, and reservoir storage (RS) from 1998 to 2022 by calculating annual means and applying the Mann-Kendall trend test for each site using the pyMannKendall Python package¹²². For sites that have statistically significant (p - value < 0.05) trends, we calculated the mean Sen's slope and p-values across sites. Additionally, we calculated the difference between the 3-year reference period for the 2012, 2018, and 2020–2021 droughts relative to the 2001–02 droughts to assess if there were any long-term shifts in the baseline values.

To estimate the changes in reference year values due to the drought episodes, we calculated the differences between pre- and post-event baselines for P, AT, RE, Q, WT, SC, and RS. We then identified the drought recovery time for each response variable (Q, WT, SC) as the first year when each variable returned to (or exceeded) the pre-drought reference baseline (3 years preceding the drought episode). Drought recovery was not calculated for the 2020–21 drought due to insufficient post-event data. Further, sites that did not have data for every year between the end of the drought episode (i.e. 2003, 2013, 2019) and the end of the record (2022) were removed for this analysis.

To further determine the extent of drought impacts and recovery, we analyzed whether precipitation-runoff relationships were altered during drought episodes following the methodology in Saft et al. (2015)²¹ and several others²¹, with slight modifications to align with the rest of our workflow. Briefly, we used the calculated annual runoff and precipitation, utilizing our previous data criteria to exclude years with missing data instead of that used in ref.²¹. We then applied a Box-Cox transform to annual runoff, and assumed annual precipitation data followed an approximate normal distribution²¹. While Saft et al. (2015)²¹ define drought years based on rainfall anomalies, we used our drought events identified using SPEI (Section 5.4) to account for the importance of warming air temperatures in the UCRB. Given the relatively short span of our analysis (1998–2022), we also analyzed drought years from all episodes together rather than separate dry periods for a given site.

Following Saft et al. (2015)²¹, we applied Equation (3) to identify whether the precipitation-runoff relationship differs in drought years relative to reference years for each site:

$$R = a_0 + a_1 I + a_2 P + \epsilon \quad (3)$$

where R is the Box-Cox transformed annual runoff, P is annual precipitation, I is the drought year indicator (where 1 represents a drought year and 0 represents a reference year), a_0 is the intercept, a_1 and a_2 are the regression coefficients and ϵ are the residuals for the regression. Autocorrelation of the residuals was computed using the method from Wallis and O'Connell (1972)¹²³. The regression was refitted after subtracting the product of the autocorrelation and the value of the variable from the previous timestep from the current timestep, following Haan (2002)¹²⁴.

Catchment attributes affecting spatial variability of drought response

Catchment attributes such as climate, geology, land use, vegetation, and anthropogenic activities influence hydrological variables and their response to disturbance. We identified attributes that make catchments more vulnerable or resilient to drought using different methods described below. The watershed response to drought is based on their relative change of each variable (Q, WT, SC), as described in Section 5.5. For all analyses, we report results for both individual attributes and aggregated attributes categories (SI Table S.12). Attribute categories were identified following Ciulla et al. (2024)¹²⁵ using a network of attributes in which each node represents an attribute and edge weights encode the extent of similarity between attributes (SI Fig. S.25). A community detection algorithm was used to cluster attributes into a small number of interpretable categories. For example, the temperatures category contains attributes like the basin average air temperature and monthly average air temperature. Other categories may have resulted due to complex physical relationships such as baseflow index and aspect.

Spearman correlations were calculated between each catchment attribute and the relative change for each variable (Q, WT, and SC). Correlations with p-values less than 0.05 were considered statistically significant. Additionally, we identified the predominant attributes of two groups of catchments that exhibited either positive or negative relative changes for each variable during drought. For this, attribute z-scores were computed as the difference between the group mean and the overall mean, normalized by the standard deviation across all catchments for that variable¹²⁵. For these analyses, aggregated categories values were obtained by averaging across all attributes within each category.

We also identified attributes influencing streamflow changes during drought episodes using an XGBoost machine learning regression model¹²⁶. The model predicts median monthly relative changes in Q, averaged across all available drought events, using the 250 GAGES-II catchment attributes as input variables (Fig. 5D). This analysis could not be done for WT and SC due to lack of target data.

The XGBoost model was applied to 165 out of 202 catchments that had streamflow observations, with 35 catchments excluded as outliers (monthly median relative changes greater than ± 3 standard deviations from the mean). Removing these 35 catchments did not substantially change the attribute distributions (all changes were less than 0.6 standard deviations from the mean). We also verified that the removal of outliers did not meaningfully affect the feature importance calculations. The XGBoost model used 100 estimators and was optimized for best model performance using the root mean squared errors. Model hyperparameters were tuned using the GridSearch algorithm in sci-kit learn with 10 repetitions using a 5-fold cross-validation for each of the 729 unique combination of hyperparameters, for a total of 36,450 trials (SI Table S.13).

To reduce the number of input features, and particularly redundant attributes in the GAGES-II dataset¹²⁵, we used a recursive feature elimination (RFE) method with cross validation. Attributes were eliminated according to their importance scores, computed using two different methods: (a) permutation feature importances and (b) Shapley values, calculated using the respective functions in scikit-learn. The permutation feature importance measures the decrease in model performance when values of input features are randomly shuffled. A higher drop in performance corresponds to higher feature importance. This model-agnostic method captures the real contribution of each feature, including interactions with other features, and gives a more realistic and interpretable estimate of how much each variable actually affects predictions in practice¹²⁷. Shapley values are based on cooperative game theory and are computed by assigning a contribution value to each feature by evaluating its average marginal impact across all possible combinations of features¹²⁸. Features were successively eliminated based on the average importance of an ensemble of 30 independent model runs using 80/20 train-test splits. Ten independent XGBoost models (each averaging 30 trials computing feature importances with different random seeds) were trained for the RFE. The lowest-ranking input

features were dropped with each iteration (10 when more than 100 features were present, and 5 for 50–100 features). When fewer than 50 features remained, we trained 50 models and eliminated only the single lowest-ranking feature.

The optimal set of attributes identified from the RFE was used to train a new XGBoost model following the same hyperparameter optimization method, which resulted in minor changes (SI S.13). We then performed 100 independent XGBoost runs using 80/20 train-test splits, calculating their feature importance over 30 randomly seeded trials. The ensemble runs provide robustness to variations in choice of train-test splits. The final feature importance scores were normalized to sum to 1, and indicate the predominant attributes influencing relative changes of Q across all drought episodes. Aggregated category values were calculated by adding the feature importance scores of individual attributes.

Data availability

The daily observations of streamflow, water temperature, and specific conductance used in this study are publicly available from USGS NWIS at <https://waterdata.usgs.gov/nwis>. The GAGES-II dataset is publicly available and can be accessed at <https://pubs.er.usgs.gov/publication/70046617>. All input, processed data used in this study, and outputs are publicly available with a CCBy4 license on the U.S. Department of Energy Environmental Systems Science Data Infrastructure for a Virtual Ecosystem (ESS-DIVE) data repository at <https://doi.org/10.15485/255189472>.

Code availability

All code used in this study is publicly available with a CCBy4 license on the U.S. Department of Energy Environmental Systems Science Data Infrastructure for a Virtual Ecosystem (ESS-DIVE) data repository at <https://doi.org/10.15485/255189472>.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Charuleka Varadharajan.

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Author contributions

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